

Name: Solutions

Dolphin ID: _____

Midterm policies

- The only resource you are allowed to use on this exam is a single one-sided page of notes (8.5×11 inches).
- You may only have a writing utensil, a drink, and a small, *quiet* snack (if necessary) on your desk.
- Multiple choice questions have one correct answer. Circle the letter corresponding to your selection.
- For questions that are not multiple choice, make sure to be clear in your response. For example, if you are asked to identify variables and their types,
 - don't write "continuous numeric, ordinal categorical",
 - do write "Income is a continuous numeric variable, employment status is an ordinal categorical variable".
- Double check your responses and make sure you answered all parts to each question.
- You will receive points only for legible (readable) answers.
- You have 75 minutes for this exam. You may submit your midterm and leave class *quietly* if you finish early.
- Take a breath, this midterm is to assess how well you are achieving the student learning outcomes, it is not meant to trick you. You've got this!

Question	Points Possible	Points Earned
1	16	
2	8	
3	29	
4	8	
5	8	
Total	69	

Question 1

Consider a study that wants to investigate the proportion of people that experience sleep difficulties after consuming energy drinks, π . They collected a sample among people aged 12-24 and found 365 of the 1,516 respondents reported sleeping difficulties after consuming energy drinks.

Question 1a**[2 Points]**

We want to test if the population proportion is more than 40%. What should our hypotheses be?

a. $H_0 : \pi > 0.4, H_A : \pi \leq 0.4$

c. $H_0 : \pi \leq 0.4, H_A : \pi = 0.4$

b. $H_0 : \pi \leq 0.4, H_A : \pi > 0.4$

d. $H_0 : \pi = 0.4, H_A : \pi \neq 0.4$

Question 1b**[14 Points]**

For each concept on the left identify the corresponding item on the right. Not all items on the right have a concept present.

- d 1. population parameter
g 2. sample statistic / point estimate
f 3. sample size
b 4. population
c 5. representation of uncertainty
e 6. frequentist model for π
a 7. Bayesian model for π

- ~~a.~~ $\pi|Y \sim \text{Beta}(\alpha, \beta)$, with $Y = \text{data}$
~~b.~~ people aged 12-24 that consume energy drinks
~~c.~~ credible or confidence interval
~~d.~~ π
~~e.~~ $p \sim \text{approximately } N(\pi, \frac{\pi(1-\pi)}{n})$
~~f.~~ 1516
~~g.~~ 365 / 1516
~~h.~~ 365 + 1516

Question 2**[8 Points]**~~commit~~

style guide

~~clone~~~~Quarto document~~

push

~~repository / repo~~

Match each definition (below) to a term (above) by writing the term next to its definition. Not all terms have a definition present.

Commit A “snapshot” or save point of changes made to files within a repository.

clone Download all the elements of a repo available on GitHub at that specific time.

repo Contains all of your project’s files as well as each file’s revision history.

Quarto doc Reproducible file that can contain and render both code and text.

Question 3

It is well studied that not getting enough sleep is related to worse performance. Inadequate sleep is especially dangerous for high stakes situations, like surgery. It is also likely that hospital personnel become progressively fatigued as the workweek progresses from Monday to Friday.

The data frame `surgery` contains data for elective general surgeries, to study the hypothesis that the risk of 30-day mortality (death of patient within 30 days of surgery) increases from Monday to Friday.

Rows: 32,001

Columns: 6

```
$ age          <dbl> 67.8, 39.5, 56.5, 71.0, 56.3, 57.7, 56.6, 64.2, 66.2, 20.1~
$ hour         <dbl> 9.03, 18.48, 7.88, 8.80, 12.20, 7.67, 9.53, 7.52, 16.35, 1~
$ day_of_week  <fct> Mon, Wed, Fri, Wed, Thu, Thu, Tue, Thu, Wed, Tue, Thu, Thu~
$ month        <chr> "Nov", "Sep", "Aug", "Jun", "Aug", "Dec", "Apr", "Sep", "A~
$ moonphase    <chr> "Full Moon", "New Moon", "Full Moon", "Last Quarter", "Las~
$ mortality30  <chr> "No", "No", "No", "No", "No", "No", "No", "No", "No", "No"~
```

Question 3a

[5 Points]

The variable `hour` records the hour of the day that the surgery was performed, with surgeries occurring from hour 6 to hour 19. Complete the code to define a new variable called `time_period`, that takes values: “Morning” for hours in [6,10), “Midday” for hours in [10, 14), “Afternoon” for hours in [14, 19].

```
surgery <- surgery |>
  mutate(
    time_period = case_when (
      hour ≥ 6 & hour < 10 ~ "Morning",
      hour ≥ 10 & hour < 14 ~ "Midday",
      hour ≥ 14 & hour ≤ 19 ~ "Afternoon"
    )
  )
```

Question 3b

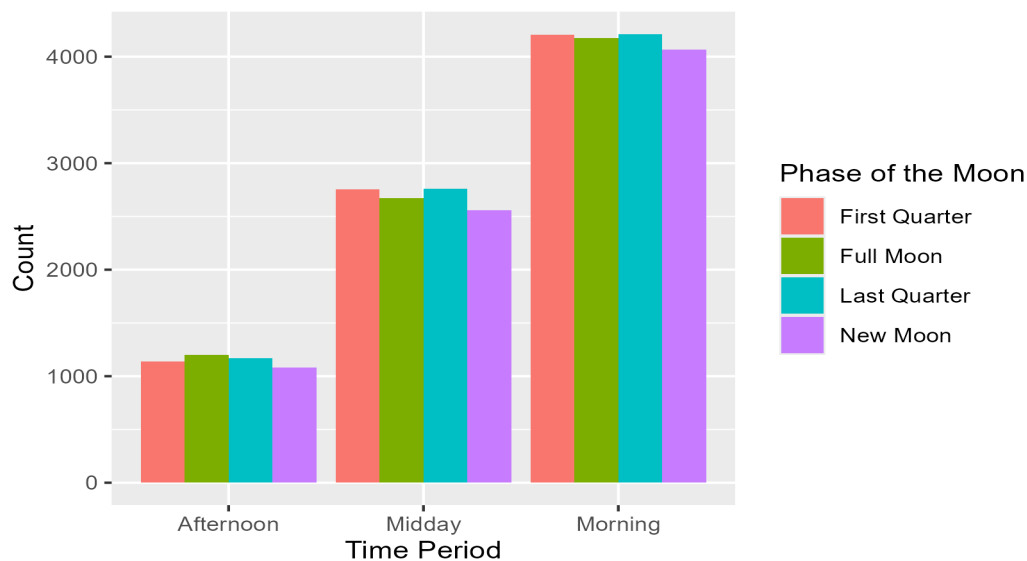
[5 Points]

Days in winter have less sunlight. Write the code to exclude surgeries from the months of December, January, and February. Save your changes back into the `surgery` data frame.

```
surgery <- surgery |>
  filter(month != "Dec" | month != "Jan" | month != "Feb")
```

Question 3c

[6 Points]



Write the code to produce the above plot.

```
ggplot(surgery,
  aes(x=time-period, fill=moonphase)) +
  geom_bar(position="dodge")
```

Question 3d

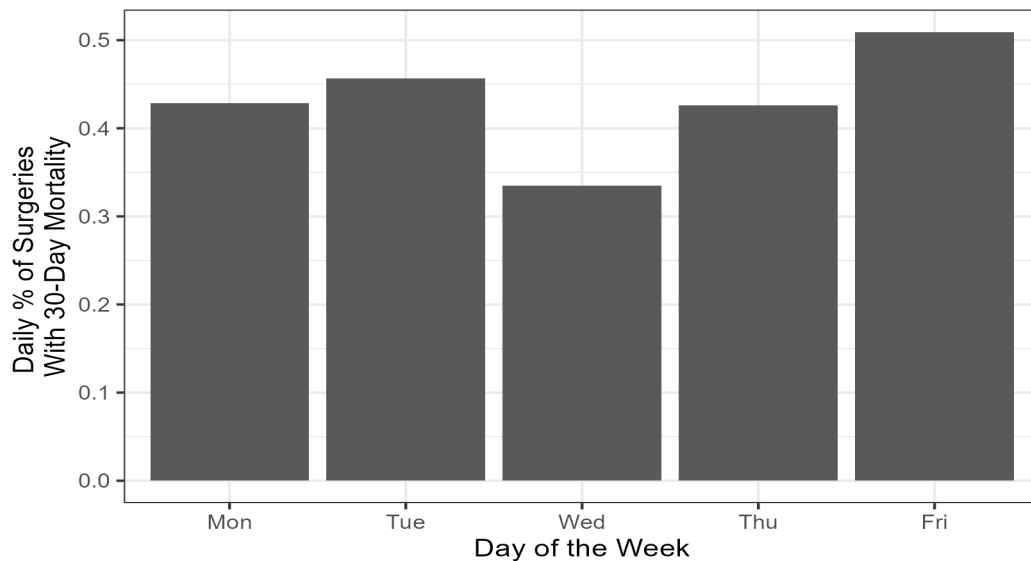
[4 Points]

Write code to drop `moon_phase` variable from the data frame. Save this change back into the `surgery` data frame.

```
surgery <- surgery |>
  select(-moonphase)
```

Question 3e

[4 Points]



Consider this alternate text for the bar plot above:

“The bar plot shows percentage of 30-day mortality on the y-axis for each day of the week on the x-axis.”

What should be added to the alt text to improve it?

Lots of different answers could be acceptable. The idea is that we should provide more information. For example: The 30-day mortality increases from Monday to Tuesday, decreases on Wednesday, and increases into Friday.

Question 3f

[3 Points]

The hypothesis was that the frequency of mortality within 30 days of surgery would increase through the workweek. Visually inspect the bar plot above in question 1e. Does that trend appear to be true in this data? Explain your reasoning.

No, there is a decrease in the middle of the week.

Question 3g

[2 Points]

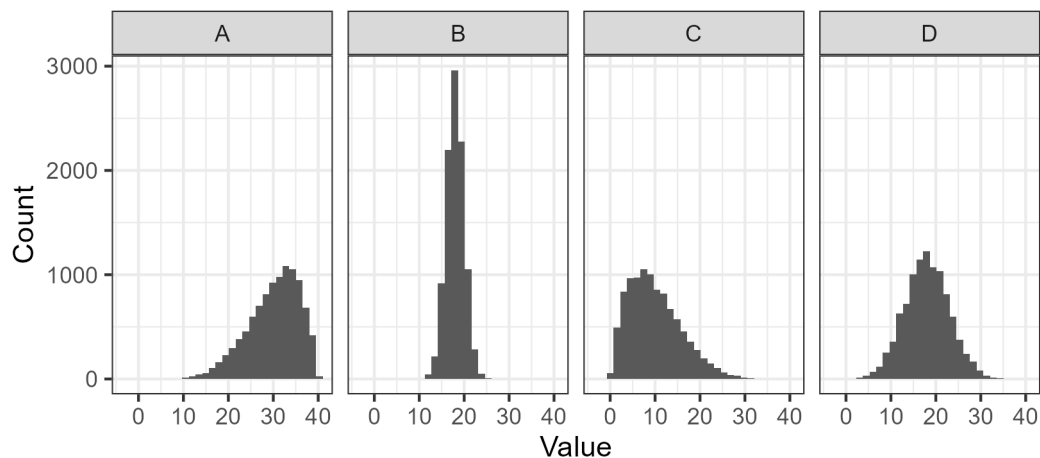
We want to get the average patient age for surgeries with mortality within 30 days, for each day of the week. Complete the code below to do this.

```
surgery |>
  group_by_____(day_of_week) |>
  summarize(mean(age))
```

```
# A tibble: 5 x 2
  day_of_week `mean(age)`
  <fct>       <dbl>
1 Mon         57.7
2 Tue         57.5
3 Wed         58.3
4 Thu         57.6
5 Fri         57.2
```

Question 4

[8 Points]



Match each row of the summary table (below) to its corresponding histogram (above) by writing the letter of the histogram in the first column.

Histogram	1st Quartile	Median	Mean	3rd Quartile	Standard Deviation
D	14.677108	17.967790	17.999224	21.37335	5.006454
A	26.365070	30.873407	30.028197	34.45612	5.718446
B	16.648853	17.978797	17.994445	19.32720	1.980910
C	5.444126	9.098417	9.974962	13.61271	5.790553

Question 5**[8 Points]**`employees`

	employee_id	name	department_id
1	1	Alice	10
2	2	Bob	20
3	3	Charlie	10
4	4	David	30

`departments`

	department_id	department_name	location
1	10	Sales	New York
2	20	Marketing	Los Angeles
3	40	Engineering	Chicago

You have two data frames: `employees` and `departments`.

What would be the output of the following code? You should show each row, column, and cell clearly.

`left_join(employees, departments)`

employee_id	name	department_id	department_name	location
1	Alice	10	Sales	New York
2	Bob	20	marketing	Los Angeles
3	Charlie	10	Sales	New York
4	David	30	NA	NA